

Discrete Mathematics

CS1111

Lecture 5: Counting

Counting

Permutations and Combinations

Permutations

- A permutation is an ordered arrangement of objects.
- The number of permutations of r distinct objects chosen from n distinct objects is denoted $P(n,r)$.
- Let $X = \{a, b, c, d\}$, 3-permutations of X :
 - $abc, abd, acb, acd, adb, adc, bac, bad, bca, bcd, bda, bdc, cab, cba, cad, cbd, cda, cdb, dab, dac, dba, dbc, dca, \& dcb$.

Permutations

- To select one letter out of 4 there are 4 options, then select one letter out of the remaining 3 letters, then select one out of the remaining 2 letters.

4 ways to fill first position	3 ways to fill second position	2 ways to fill third position
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- Total $4 \times 3 \times 2 = 24$ options

Permutations

- To calculate $P(n, r)$ use the product rule

$$\begin{aligned} P(n, r) &= n \times (n - 1) \times (n - 2) \times \dots \\ &\quad (n - r + 1) = n! / (n - r)! \end{aligned}$$

$$\begin{aligned} \text{where } n! &= n \times (n - 1) \times (n - 2) \times \dots \\ &\quad \times 3 \times 2 \times 1 \end{aligned}$$

$$n \in \mathbb{N}, \text{ and } 0 \leq r \leq n$$

Problems

- In how many ways can a group of 5 students sit in a lab of 20 computers?
- How many different car plates if a plate contains a sequence of 3 Arabic letters followed by a sequence of 4 digits which cannot be all zeros

Problems

- A group of 12 students are having lectures in a classroom that contains 20 desks
 - How many ways can the students be seated
 - How many lectures are needed so that each student can use every desk

Problems

- A menu of drinks includes: tea, plain water, orange juice and coffee. You can drink 3 times a day, each time you can select one drink, how many ways to select a daily list of drinks if repetition is allowed/not allowed?

Relations

- Let $X = \{a, b, c\}$
- $P(n, 1) = n! / (n-1)! = n \cdot (n-1)! / (n-1)! = n$
 - 1-permutations of X : a, b, c
- $P(n, n) = n! / (n-n)! = n! / 0! = n! / 1 = n!$
 - 3-permutations of X : $abc, acb, bac, bca, cab, cba$ $3! = 3 \cdot 2 \cdot 1 = 6$
- $P(n, n-1) = n! / (n-n+1)! = n! / (1)! = n!$
- $P(n, 0) = n! / (n-0)! = n! / n! = 1$

Problems

- How many ways to select a football team from a group of 14 players if
 - Any player can play at any position
 - Of the 14 players, there are two goal keepers
 - There is one goal keeper

Combinations

- A combination is an unordered selection of elements from some set.
- ORDER does NOT matter
 - $abc = bca = acb = \dots$
- The number of combinations of r distinct objects chosen from n distinct objects is denoted by $C(n,r)$ or nCr , and is read “ n chosen r .”

$$C(n,r) = P(n,r)/r! = n! / ((n-r)!r!)$$

Combinations

- Since the order does not matter in combinations, there are fewer combinations than permutations. The combinations are a "subset" of the permutations.
- $C(n,r) = P(n,r)/r! = n! / ((n-r)!r!)$

$P(n, r)$ vs $C(n, r)$

- Let $X = \{a, b, c, d\}$, subsets of X of 3 elements
 - $\{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}$.
 - 3-combinations of X
 - $C(4, 3) = P(4, 3)/3! = 24/6 = 4$

Recall

- 3-permutations of X :
 - $abc, abd, acb, acd, adb, adc, bac, bad, bca, bcd, bda, bdc, cab, cba, cad, cbd, cda, cdb, dab, dac, dba, dbc, dca, \& dcb$.

Problems

- How many ways to select a football team from a group of 14 players if
 - Any player can play at any position
 - Order does NOT matter so $c(14, 11) = 14! / ((14-11)!(11)!)$
 $= 14! / (3! 11!) = 14 * 13 * 12 / 3! = 14 * 13 * 6 * 2 / 6 =$
 $14 * 13 * 2 = 182 * 2 = 364$
 - Of the 14 players, there are two goal keepers
 - $C(2, 1) * (12, 10) =$
 - There is one goal keeper
 - $C(1, 1) * C(13, 10) =$

Relations

- $C(n, 1) = P(n, 1) / 1! = P(n, 1) = n$
 - 30 ways to select 1 student from a group of 30 students
- $C(n, n) = P(n, n) / n! = n! / n! = 1$
 - Only one way to select 30 students from a group of 30 students
- $C(n, 0) = P(n, 0) / 0! = 1 / 0! = 1 / 1 = 1$
 - To select no student from a group of 30 students, the only way to do nothing!

Relations

- $C(n, r) = C(n, n - r)$
 - $C(n, r) = n! / ((n - r)! r!)$
 - $C(n, n-r) = n! / ((n - (n - r))! (n-r)!) =$
 $n! / ((n - n + r)! (n - r)!) = n! / (r! (n - r)!) =$
 $C(n, r)$

Problem

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- $P(30, 3) = 30!/(30 - 3)! = 30 * 29 * 28$

Problem

- How many permutations of $\{a, b, c, d, e, f, g\}$ end with a?
- Note that the set has 7 elements
- The last character must be a
- The rest can be in any order
- Thus, we want a 6-permutation on the set $\{b, c, d, e, f, g\}$
- $P(6,6) = 6! = 720$

Problem

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$$- C(5, 3) = 5! / ((5 - 3)! (3)!) = 5! / 2!3! = 10$$

Problem

- What is the number of the possible permutations of the word “problem”?

- Let n be a nonnegative integer, then

$$\sum_{k=0}^{k=n} C(n, k) = 2^n$$

- If $|X| = n$, then $C(n, k)$ is the number of subsets of X of k elements, the above formula is the number of all subsets of X , that is $|P(X)|$ which is $= 2^n$.

Problem

- How many ways to select a committee of one or two member out of 8 persons?